DOKUMEN PERANCANGAN SISTEM & PAPER PROJECT

SHOE WORKSHOP INDONESIA • DIVISI IT & DATA ANALYST

PERANCANGAN SKEMA DATABASE MANAJEMEN BRAND (B2B) & PROGRESS TRACKING

Arsitektur Skema Relasional 12 Tabel Standalone, Lifecycle Unit Tracking, Append-Only Progress Log, & Public Token Link

NOMOR DOKUMEN PAPER

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STATUS PERANCANGAN

✅ **Final Draft Approved (Zero Database Breaking Changes)**

DIVISI TERKAIT

Divisi Workshop (Produksi), Divisi Finance, & Divisi IT

LEAD ARCHITECT & PENULIS

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1. Ringkasan Keputusan (Hasil Sesi Grill-Me)

#	TOPIK	KEPUTUSAN
1	**Keterikatan dengan** `work_orders`	**Full standalone.** Seluruh data brand (unit sepatu, tahapan, progress, status) berada di tabel baru. Tabel `work_orders` **tidak disentuh sama sekali** (tidak ada kolom `brand_order_id`). Stasiun/scanner brand dibuat terpisah.
2	**Model tahapan**	**Tetap & sama untuk semua unit:** `PREPARATION → SORTIR → PRODUCTION → QC` (hardcode sebagai PHP Enum, bukan tabel master).
3	**Kedalaman tracking**	**Riwayat penuh (event log append-only):** `brand_unit_progress` (1 baris per unit per tahap per percobaan) + `brand_unit_events` (log tiap aksi).
4	**QC gagal**	**Rework loop:** QC reject → unit kembali ke tahap tertentu, `attempt_no` bertambah, riwayat percobaan lama tetap tersimpan, alasan reject tercatat (kode + catatan + foto opsional).
5	**Progress agregat batch**	**Counter denormalisasi** di `brand_orders` + `status` / `current_stage` di `brand_units`, diperbarui service tiap event; dapat direkonsiliasi ulang dari log.
6	**Visibilitas ke brand**	**Public tracking link bertoken** (read-only, tanpa login) + tabel `brand_batch_updates` untuk catatan/foto update.
7	**Dispatch**	**Parsial didukung:** `brand_dispatches` + `brand_dispatch_items`; unit berstatus `DISPATCHED` sendiri-sendiri; batch `PARTIALLY_DISPATCHED` → `DISPATCHED`.
8	**Foto unit**	**Foto per unit per tahap:** tabel `brand_unit_photos` bertipe `BEFORE / PROCESS / AFTER / REJECT`. `BEFORE` wajib saat tahap Preparation selesai, `AFTER` wajib saat QC lolos, `REJECT` wajib saat QC menolak. Visibilitas publik diatur per foto.
9	**Invoice & harga**	**Ditunda.** Belum ada konsep; skema ini sengaja tidak memuat harga/invoice dan akan ditambah lewat migration terpisah nanti.
10	**Data lama**	**Tidak ada** data brand di `work_orders`, sehingga tidak diperlukan migrasi/konversi data historis.

Konsekuensi Utama dari Keputusan "Full Standalone"

> [!IMPORTANT]
>
> - Isolasi dari rak pickup/kasir CS menjadi **otomatis dan permanen** — unit brand tidak berada di `work_orders`, sehingga status khusus `BRAND_FINISHED` tidak lagi perlu "disaring" dari query CS.
> - Scanner, halaman mobile teknisi, dan cetak SPK **harus dibuat paralel** untuk brand (rute `/m/brand/...`), tidak dapat memakai ulang `ProductionControlling` / `SpkTracker` begitu saja (hanya pola dan trait-nya yang bisa dicontoh).
> - Laporan/dashboard/keuangan eksisting yang membaca `work_orders` **tidak akan melihat data brand**. Perlu keputusan terpisah untuk laporan & penagihan brand (lihat Bab 10).

🎯 Tujuan Perancangan Skema Database

Perancangan arsitektur dan sistem ini disusun secara spesifik untuk mencapai sasaran strategis dan operasional bagi **Divisi Workshop (Produksi), Divisi Finance, & Divisi IT** sebagai berikut:

1. Merancang struktur 12 tabel database relasional baru yang modular dan berdiri sendiri untuk kebutuhan kemitraan brand B2B dari tahap registrasi hingga penagihan.
2. Mendukung pelacakan status bertingkat (*Lifecycle Unit Tracking & Append-Only Progress Logs*) untuk merekam jejak pengerjaan stasiun sol, upper, jahit, dan finishing.
3. Menyediakan token akses unik (`tracking_token`) pada setiap batch order untuk memfasilitasi live tracking publik tanpa memerlukan akun login pabrik.
4. Menjamin nol dampak kerusakan (*zero breaking changes*) terhadap tabel transaksi workshop eksisting (`work_orders`, `customers`, `order_payments`).

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2. Entity Relationship Diagram

```
erDiagram
    brand_partners ||--o{ brand_orders : "memiliki batch"
    users ||--o{ brand_orders : "created_by"
    brand_orders ||--|{ brand_order_items : "terdiri dari model"
    brand_orders ||--o{ brand_units : "memuat unit"
    brand_order_items ||--o{ brand_units : "menurunkan unit"
    brand_units ||--o{ brand_unit_progress : "tahapan x percobaan"
    brand_units ||--o{ brand_unit_events : "log aksi"
    brand_unit_progress ||--o{ brand_unit_events : "sumber event"
    brand_unit_progress ||--o{ brand_qc_rejections : "QC gagal"
    brand_orders ||--o{ brand_batch_updates : "update untuk brand"
    brand_orders ||--o{ brand_dispatches : "surat jalan"
    brand_dispatches ||--|{ brand_dispatch_items : "isi"
    brand_units ||--o| brand_dispatch_items : "dikirim"
    users ||--o{ brand_unit_progress : "teknisi"
    brand_units ||--o{ brand_unit_photos : "foto unit"
    brand_unit_progress ||--o{ brand_unit_photos : "foto tahap"
    brand_qc_rejections ||--o{ brand_unit_photos : "foto reject"
    users ||--o{ brand_unit_photos : "uploaded_by"
```

Total 12 tabel baru, 0 perubahan pada tabel eksisting.

NO.	TABEL	FUNGSI	SIFAT DATA
1	`brand_partners`	Master mitra brand	Master
2	`brand_orders`	Batch induk + counter progress + token publik	Transaksi + Agregat
3	`brand_order_items`	Kartu model sepatu per batch (size matrix)	Transaksi
4	`brand_units`	1 baris per pasang sepatu (barcode satuan)	Transaksi + Snapshot status
5	`brand_unit_progress`	1 baris per unit × tahap × percobaan	Transaksi
6	`brand_unit_events`	Log aksi append-only (mulai/jeda/lanjut/selesai/reject)	Log (immutable)
7	`brand_qc_rejections`	Detail alasan QC reject & tujuan rework	Transaksi
8	`brand_batch_updates`	Catatan/foto update yang tampil ke brand	Transaksi
9	`brand_dispatches`	Surat jalan pengembalian (header)	Transaksi
10	`brand_dispatch_items`	Unit yang dikirim pada tiap surat jalan	Transaksi
11	`brand_number_sequences`	Penomoran batch harian anti-duplikasi	Utilitas
12	`brand_unit_photos`	Foto BEFORE/PROCESS/AFTER/REJECT per unit & tahap	Transaksi

3. Spesifikasi Tabel (DDL)

3.1 `brand_partners`

```
CREATE TABLE `brand_partners` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `code` VARCHAR(50) NOT NULL UNIQUE COMMENT 'BRD-VNTL, BRD-BRDO',
    `name` VARCHAR(255) NOT NULL,
    `pic_name` VARCHAR(255) NOT NULL,
    `pic_phone` VARCHAR(50) NOT NULL,
    `pic_email` VARCHAR(255) NULL,
    `address` TEXT NULL,
    `logo_path` VARCHAR(255) NULL,
    `tier_level` VARCHAR(50) NOT NULL DEFAULT 'STANDARD' COMMENT 'STANDARD | EXCLUSIVE | VIP',
    `status` VARCHAR(20) NOT NULL DEFAULT 'ACTIVE' COMMENT 'ACTIVE | INACTIVE',
    `notes` TEXT NULL,
    `created_at` TIMESTAMP NULL,
    `updated_at` TIMESTAMP NULL,
    `deleted_at` TIMESTAMP NULL,
    INDEX `idx_brand_partners_status` (`status`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

3.2 `brand_orders` (Batch Induk + Counter Agregat)

```
CREATE TABLE `brand_orders` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `batch_number` VARCHAR(64) NOT NULL UNIQUE COMMENT 'BR-YYMM-DD-XXXX',
    `brand_partner_id` BIGINT UNSIGNED NOT NULL,
    `created_by` BIGINT UNSIGNED NOT NULL,
    `project_name` VARCHAR(255) NOT NULL,
    `po_reference_no` VARCHAR(100) NULL,
    `received_date` DATE NOT NULL,
    `deadline_date` DATE NULL,
    `special_notes` TEXT NULL,

    -- STATUS BATCH
    `status` VARCHAR(30) NOT NULL DEFAULT 'DRAFT'
        COMMENT 'DRAFT | INBOUND | IN_PRODUCTION | FINISHED | PARTIALLY_DISPATCHED |
DISPATCHED | CANCELLED',

    -- COUNTER DENORMALISASI (diupdate BrandTrackingService, bisa direkonsiliasi)
    `total_units` INT UNSIGNED NOT NULL DEFAULT 0,
    `units_waiting` INT UNSIGNED NOT NULL DEFAULT 0,
    `units_in_progress` INT UNSIGNED NOT NULL DEFAULT 0,
    `units_rework` INT UNSIGNED NOT NULL DEFAULT 0,
    `units_finished` INT UNSIGNED NOT NULL DEFAULT 0 COMMENT 'lolos QC, belum dispatch',
    `units_dispatched` INT UNSIGNED NOT NULL DEFAULT 0,
    `progress_pct` DECIMAL(5,2) NOT NULL DEFAULT 0 COMMENT '(finished + dispatched) / total x
100',
    `stage_counts` JSON NULL COMMENT '{"PREPARATION":{"waiting":0,"active":0,"done":0}, ...}
untuk deteksi bottleneck',
    `counters_synced_at` TIMESTAMP NULL COMMENT 'terakhir direkonsiliasi penuh dari log',

    -- TRACKING PUBLIK UNTUK BRAND
    `public_token` VARCHAR(64) NULL UNIQUE COMMENT 'token link /brand-report/{token}',
    `public_enabled` TINYINT(1) NOT NULL DEFAULT 1,

    -- MILESTONE WAKTU
    `production_started_at` TIMESTAMP NULL,
    `finished_at` TIMESTAMP NULL COMMENT 'seluruh unit lolos QC',
    `dispatched_at` TIMESTAMP NULL COMMENT 'seluruh unit terkirim',
    `created_at` TIMESTAMP NULL,
    `updated_at` TIMESTAMP NULL,
    `deleted_at` TIMESTAMP NULL,

    INDEX `idx_brand_orders_status` (`status`),
    INDEX `idx_brand_orders_partner` (`brand_partner_id`, `status`),
    INDEX `idx_brand_orders_deadline` (`deadline_date`),
    FOREIGN KEY (`brand_partner_id`) REFERENCES `brand_partners`(`id`) ON DELETE RESTRICT,
    FOREIGN KEY (`created_by`) REFERENCES `users`(`id`) ON DELETE RESTRICT
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

3.3 `brand_order_items` (Kartu Model / Smart Size Matrix)

```
CREATE TABLE `brand_order_items` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `brand_order_id` BIGINT UNSIGNED NOT NULL,
    `line_no` SMALLINT UNSIGNED NOT NULL COMMENT 'urutan kartu model dalam batch',
    `model_name` VARCHAR(255) NOT NULL,
    `color` VARCHAR(100) NULL,
    `service_name` VARCHAR(255) NOT NULL COMMENT 'mis. Sol Reglue + Jahit',
    `service_detail` TEXT NULL,
    `size_breakdown` JSON NOT NULL COMMENT '{"39":10,"40":10,"41":10}',
    `quantity` INT UNSIGNED NOT NULL,
    `unit_seq_start` INT UNSIGNED NOT NULL COMMENT 'nomor urut flat awal',
    `unit_seq_end` INT UNSIGNED NOT NULL COMMENT 'nomor urut flat akhir',
    `created_at` TIMESTAMP NULL,
    `updated_at` TIMESTAMP NULL,
    UNIQUE KEY `uq_brand_items_line` (`brand_order_id`, `line_no`),
    FOREIGN KEY (`brand_order_id`) REFERENCES `brand_orders`(`id`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

3.4 `brand_units` (1 Baris = 1 Pasang Sepatu)

```
CREATE TABLE `brand_units` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `brand_order_id` BIGINT UNSIGNED NOT NULL,
    `brand_order_item_id` BIGINT UNSIGNED NOT NULL,
    `unit_seq` INT UNSIGNED NOT NULL COMMENT 'nomor urut flat 1..N melintasi seluruh model',
    `unit_code` VARCHAR(80) NOT NULL UNIQUE COMMENT 'BR-2609-30-0001-01 (isi barcode)',
    `shoe_size` VARCHAR(20) NOT NULL,

    -- SNAPSHOT STATUS (untuk query cepat; kebenaran historis ada di progress/events)
    `status` VARCHAR(20) NOT NULL DEFAULT 'WAITING'
        COMMENT 'WAITING | IN_PROGRESS | PAUSED | REWORK | FINISHED | DISPATCHED',
    `current_stage` VARCHAR(20) NOT NULL DEFAULT 'PREPARATION'
        COMMENT 'PREPARATION | SORTIR | PRODUCTION | QC | DONE',
    `current_attempt_no` TINYINT UNSIGNED NOT NULL DEFAULT 1,
    `rework_count` TINYINT UNSIGNED NOT NULL DEFAULT 0,
    `current_location` VARCHAR(255) NOT NULL DEFAULT 'Gudang Inbound',
    `notes` TEXT NULL,

    `started_at` TIMESTAMP NULL COMMENT 'tahap pertama dimulai',
    `qc_passed_at` TIMESTAMP NULL,
    `dispatched_at` TIMESTAMP NULL,
    `created_at` TIMESTAMP NULL,
    `updated_at` TIMESTAMP NULL,

    UNIQUE KEY `uq_brand_units_seq` (`brand_order_id`, `unit_seq`),
    INDEX `idx_brand_units_status` (`brand_order_id`, `status`),
    INDEX `idx_brand_units_stage` (`brand_order_id`, `current_stage`, `status`),
    FOREIGN KEY (`brand_order_id`) REFERENCES `brand_orders`(`id`) ON DELETE CASCADE,
    FOREIGN KEY (`brand_order_item_id`) REFERENCES `brand_order_items`(`id`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

3.5 `brand_unit_progress` (Unit × Tahap × Percobaan)

```
CREATE TABLE `brand_unit_progress` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `brand_unit_id` BIGINT UNSIGNED NOT NULL,
    `brand_order_id` BIGINT UNSIGNED NOT NULL COMMENT 'denormalisasi untuk agregasi cepat per
batch',
    `stage` VARCHAR(20) NOT NULL COMMENT 'PREPARATION | SORTIR | PRODUCTION | QC',
    `attempt_no` TINYINT UNSIGNED NOT NULL DEFAULT 1,
    `status` VARCHAR(20) NOT NULL DEFAULT 'PENDING'
        COMMENT 'PENDING | IN_PROGRESS | PAUSED | DONE | VOIDED',
    `technician_id` BIGINT UNSIGNED NULL,

    `started_at` TIMESTAMP NULL,
    `finished_at` TIMESTAMP NULL,
    `last_paused_at` TIMESTAMP NULL COMMENT 'jeda yang sedang berjalan',
    `pause_reason` VARCHAR(100) NULL,
    `paused_seconds` INT UNSIGNED NOT NULL DEFAULT 0 COMMENT 'akumulasi jeda selesai',
    `net_seconds` INT UNSIGNED NULL COMMENT '(finished - started) - paused; diisi saat DONE',
    `result` VARCHAR(20) NULL COMMENT 'khusus QC: PASSED | REJECTED',
    `notes` TEXT NULL,
    `created_at` TIMESTAMP NULL,
    `updated_at` TIMESTAMP NULL,

    UNIQUE KEY `uq_brand_progress_unit_stage_attempt` (`brand_unit_id`, `stage`,
`attempt_no`),
    INDEX `idx_brand_progress_batch_stage` (`brand_order_id`, `stage`, `status`),
    INDEX `idx_brand_progress_tech` (`technician_id`, `status`),
    FOREIGN KEY (`brand_unit_id`) REFERENCES `brand_units`(`id`) ON DELETE CASCADE,
    FOREIGN KEY (`brand_order_id`) REFERENCES `brand_orders`(`id`) ON DELETE CASCADE,
    FOREIGN KEY (`technician_id`) REFERENCES `users`(`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

3.6 `brand_unit_events` (Log Append-Only)

```
CREATE TABLE `brand_unit_events` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `brand_unit_id` BIGINT UNSIGNED NOT NULL,
    `brand_order_id` BIGINT UNSIGNED NOT NULL,
    `progress_id` BIGINT UNSIGNED NULL,
    `event_type` VARCHAR(30) NOT NULL
        COMMENT 'STARTED | PAUSED | RESUMED | COMPLETED | QC_PASSED | QC_REJECTED |
REWORK_STARTED | DISPATCHED | PHOTO_ADDED | NOTE | CORRECTION',
    `stage` VARCHAR(20) NULL,
    `attempt_no` TINYINT UNSIGNED NULL,
    `from_status` VARCHAR(20) NULL,
    `to_status` VARCHAR(20) NULL,
    `actor_id` BIGINT UNSIGNED NULL,
    `reason_code` VARCHAR(50) NULL,
    `notes` TEXT NULL,
    `meta` JSON NULL,
    `created_at` TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

    INDEX `idx_brand_events_unit` (`brand_unit_id`, `created_at`),
    INDEX `idx_brand_events_batch` (`brand_order_id`, `created_at`),
    INDEX `idx_brand_events_type` (`event_type`, `created_at`),
    FOREIGN KEY (`brand_unit_id`) REFERENCES `brand_units`(`id`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

> [!NOTE] Tabel ini **tidak memiliki** `updated_at` dan **tidak pernah di-UPDATE/DELETE** oleh aplikasi. Koreksi dilakukan dengan menambah event `CORRECTION`, bukan mengubah event lama.

3.7 `brand_qc_rejections`

```
CREATE TABLE `brand_qc_rejections` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `brand_unit_id` BIGINT UNSIGNED NOT NULL,
    `qc_progress_id` BIGINT UNSIGNED NOT NULL COMMENT 'baris brand_unit_progress tahap QC yang
gagal',
    `attempt_no` TINYINT UNSIGNED NOT NULL COMMENT 'percobaan yang ditolak',
    `reason_code` VARCHAR(50) NOT NULL
        COMMENT 'GLUE_WEAK | STITCH_LOOSE | COLOR_MISMATCH | DIRTY | DEFORMED | OTHER',
    `notes` TEXT NULL,
    -- foto reject disimpan di brand_unit_photos (type = REJECT, rejection_id)
    `return_to_stage` VARCHAR(20) NOT NULL COMMENT 'SORTIR | PRODUCTION (atau PREPARATION)',
    `rejected_by` BIGINT UNSIGNED NOT NULL,
    `resolved_at` TIMESTAMP NULL COMMENT 'saat attempt berikutnya lolos QC',
    `created_at` TIMESTAMP NULL,
    `updated_at` TIMESTAMP NULL,
    INDEX `idx_brand_rejections_unit` (`brand_unit_id`),
    INDEX `idx_brand_rejections_reason` (`reason_code`),
    FOREIGN KEY (`brand_unit_id`) REFERENCES `brand_units`(`id`) ON DELETE CASCADE,
    FOREIGN KEY (`qc_progress_id`) REFERENCES `brand_unit_progress`(`id`) ON DELETE CASCADE,
    FOREIGN KEY (`rejected_by`) REFERENCES `users`(`id`) ON DELETE RESTRICT
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

3.8 `brand_batch_updates`

```
CREATE TABLE `brand_batch_updates` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `brand_order_id` BIGINT UNSIGNED NOT NULL,
    `user_id` BIGINT UNSIGNED NULL,
    `title` VARCHAR(255) NOT NULL,
    `notes` TEXT NULL,
    `photo_path` VARCHAR(500) NULL,
    `is_public` TINYINT(1) NOT NULL DEFAULT 1 COMMENT 'tampil di halaman publik brand',
    `created_at` TIMESTAMP NULL,
    `updated_at` TIMESTAMP NULL,
    INDEX `idx_brand_updates_batch` (`brand_order_id`, `created_at`),
    FOREIGN KEY (`brand_order_id`) REFERENCES `brand_orders`(`id`) ON DELETE CASCADE,
    FOREIGN KEY (`user_id`) REFERENCES `users`(`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

3.9 `brand_dispatches` & `brand_dispatch_items`

```
CREATE TABLE `brand_dispatches` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `dispatch_number` VARCHAR(64) NOT NULL UNIQUE COMMENT 'SJ-BR-YYMM-DD-XXXX',
    `brand_order_id` BIGINT UNSIGNED NOT NULL,
    `dispatched_by` BIGINT UNSIGNED NOT NULL,
    `dispatched_at` TIMESTAMP NOT NULL,
    `total_units` INT UNSIGNED NOT NULL,
    `receiver_name` VARCHAR(255) NULL,
    `courier_name` VARCHAR(255) NULL,
    `vehicle_plate` VARCHAR(30) NULL,
    `signature_path` VARCHAR(500) NULL,
    `notes` TEXT NULL,
    `created_at` TIMESTAMP NULL,
    `updated_at` TIMESTAMP NULL,
    INDEX `idx_brand_dispatch_batch` (`brand_order_id`),
    FOREIGN KEY (`brand_order_id`) REFERENCES `brand_orders`(`id`) ON DELETE RESTRICT,
    FOREIGN KEY (`dispatched_by`) REFERENCES `users`(`id`) ON DELETE RESTRICT
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE `brand_dispatch_items` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `brand_dispatch_id` BIGINT UNSIGNED NOT NULL,
    `brand_unit_id` BIGINT UNSIGNED NOT NULL UNIQUE COMMENT 'satu unit hanya boleh dikirim
sekali',
    `created_at` TIMESTAMP NULL,
    FOREIGN KEY (`brand_dispatch_id`) REFERENCES `brand_dispatches`(`id`) ON DELETE CASCADE,
    FOREIGN KEY (`brand_unit_id`) REFERENCES `brand_units`(`id`) ON DELETE RESTRICT
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

3.10 `brand_number_sequences`

```
CREATE TABLE `brand_number_sequences` (
    `seq_key` VARCHAR(40) NOT NULL PRIMARY KEY COMMENT 'BATCH-260930 | DISPATCH-260930',
    `last_value` INT UNSIGNED NOT NULL DEFAULT 0,
    `updated_at` TIMESTAMP NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Penomoran memakai `SELECT ... FOR UPDATE` di dalam transaksi sehingga aman dari duplikasi saat dua admin menyimpan batch bersamaan.

3.11 `brand_unit_photos` (Foto per Unit per Tahap)

```
CREATE TABLE `brand_unit_photos` (
    `id` BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    `brand_unit_id` BIGINT UNSIGNED NOT NULL,
    `brand_order_id` BIGINT UNSIGNED NOT NULL COMMENT 'denormalisasi untuk galeri per batch',
    `progress_id` BIGINT UNSIGNED NULL COMMENT 'tahap terkait (null = foto umum unit)',
    `rejection_id` BIGINT UNSIGNED NULL COMMENT 'terisi jika type = REJECT',
    `type` VARCHAR(20) NOT NULL COMMENT 'BEFORE | PROCESS | AFTER | REJECT',
    `stage` VARCHAR(20) NULL COMMENT 'salinan tahap saat foto diambil',
    `attempt_no` TINYINT UNSIGNED NOT NULL DEFAULT 1,
    `photo_path` VARCHAR(500) NOT NULL,
    `thumb_path` VARCHAR(500) NULL,
    `caption` VARCHAR(255) NULL,
    `is_public` TINYINT(1) NOT NULL DEFAULT 0 COMMENT 'tampil di /brand-report/{token};
default BEFORE/AFTER = 1 oleh service',
    `uploaded_by` BIGINT UNSIGNED NULL,
    `taken_at` TIMESTAMP NULL,
    `created_at` TIMESTAMP NULL,
    `updated_at` TIMESTAMP NULL,
    `deleted_at` TIMESTAMP NULL,

    INDEX `idx_brand_photos_unit` (`brand_unit_id`, `type`, `attempt_no`),
    INDEX `idx_brand_photos_batch` (`brand_order_id`, `type`, `is_public`),
    FOREIGN KEY (`brand_unit_id`) REFERENCES `brand_units`(`id`) ON DELETE CASCADE,
    FOREIGN KEY (`brand_order_id`) REFERENCES `brand_orders`(`id`) ON DELETE CASCADE,
    FOREIGN KEY (`progress_id`) REFERENCES `brand_unit_progress`(`id`) ON DELETE SET NULL,
    FOREIGN KEY (`rejection_id`) REFERENCES `brand_qc_rejections`(`id`) ON DELETE SET NULL,
    FOREIGN KEY (`uploaded_by`) REFERENCES `users`(`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Aturan foto wajib (divalidasi `BrandTrackingService`):

MOMEN	TIPE FOTO	SYARAT
Menyelesaikan tahap `PREPARATION` (attempt 1)	`BEFORE`	Minimal 1 foto, jika belum ada tombol SELESAI ditolak
Aksi tengah tahap (opsional)	`PROCESS`	Tidak wajib
QC `PASSED`	`AFTER`	Minimal 1 foto pada attempt yang sama
QC `REJECTED`	`REJECT`	Minimal 1 foto, ditautkan ke `brand_qc_rejections.id`

Pengelolaan file:

- Lokasi: `storage/app/public/brand/{batch_number}/{unit_code}/{type}_{timestamp}.jpg`.
- Diubah ukuran otomatis (sisi terpanjang maks. 1600 px, kualitas JPEG ~80%) beserta thumbnail untuk galeri.
- Soft delete saja; foto `AFTER` / `REJECT` yang sudah menjadi bukti QC tidak boleh dihapus, hanya diganti dengan event `CORRECTION`.
- Setiap unggahan menulis event `PHOTO_ADDED` di `brand_unit_events`.

---

4. Sistem Tracking Progress

4.1 Tahapan Tetap (PHP Enum)

```
enum BrandStage: string {
    case PREPARATION = 'PREPARATION'; // urutan 1
    case SORTIR      = 'SORTIR';      // urutan 2
    case PRODUCTION  = 'PRODUCTION';  // urutan 3
    case QC          = 'QC';          // urutan 4
}
```

4.2 Diagram Status Unit (`brand_units.status`)

```
stateDiagram-v2
    [*] ⟶ WAITING : unit di-generate
    WAITING ⟶ IN_PROGRESS : scan + MULAI (tahap 1)
    IN_PROGRESS ⟶ PAUSED : JEDA
    PAUSED ⟶ IN_PROGRESS : LANJUT
    IN_PROGRESS ⟶ WAITING : tahap selesai, antre tahap berikutnya
    IN_PROGRESS ⟶ FINISHED : QC = PASSED
    IN_PROGRESS ⟶ REWORK : QC = REJECTED
    REWORK ⟶ IN_PROGRESS : mulai tahap tujuan rework (attempt_no + 1)
    FINISHED ⟶ DISPATCHED : masuk surat jalan
    DISPATCHED ⟶ [*]
```

`WAITING` di antara tahap berarti unit sudah menyelesaikan tahap sebelumnya dan menunggu teknisi tahap berikutnya (nilai `current_stage` sudah menunjuk tahap berikutnya).

4.3 Siklus Satu Tahap (`brand_unit_progress.status`)

```
stateDiagram-v2
    [*] ⟶ PENDING : baris dibuat saat tahap menjadi giliran unit
    PENDING ⟶ IN_PROGRESS : MULAI
    IN_PROGRESS ⟶ PAUSED : JEDA
    PAUSED ⟶ IN_PROGRESS : LANJUT
    IN_PROGRESS ⟶ DONE : SELESAI
    PAUSED ⟶ DONE : SELESAI (auto-resume)
    DONE ⟶ VOIDED : unit di-rework ke tahap sebelum baris ini
```

4.4 Aturan Bisnis Inti

#	ATURAN
1	Tahap hanya boleh dimulai jika tahap sebelumnya pada `attempt_no` yang sama berstatus `DONE` (kecuali tahap 1).
2	Satu unit hanya boleh punya **satu** baris progress aktif (`IN_PROGRESS` / `PAUSED`) pada satu waktu.
3	`net_seconds = (finished_at - started_at) - paused_seconds` (selisih UNIX timestamp, bukan objek `diff()` bertanda).
4	**Auto-resume:** menekan SELESAI saat `PAUSED` otomatis menutup jeda, menambah `paused_seconds`, lalu menyelesaikan tahap.
5	**Smart auto-assign:** `technician_id` = `Auth::id()` saat MULAI; admin/supervisor boleh memilih teknisi lain.
6	QC `PASSED` → unit `FINISHED`, `current_stage = DONE`, `current_location = 'Area Staging B2B / Kardus Master'`.
7	QC `REJECTED` wajib mengisi `reason_code` dan `return_to_stage`; sistem membuat baris `brand_qc_rejections`.
8	Unit `FINISHED` baru boleh masuk `brand_dispatch_items`; unit `DISPATCHED` bersifat final (immutable).
9	Tahap `PREPARATION` tidak dapat diselesaikan tanpa minimal 1 foto `BEFORE` pada attempt 1.
10	QC `PASSED` mensyaratkan minimal 1 foto `AFTER`; QC `REJECTED` mensyaratkan minimal 1 foto `REJECT`.

4.5 Rework Loop

Contoh unit `BR-2609-30-0001-12` ditolak QC karena lem lemah, dikembalikan ke `PRODUCTION`:

PROGRESS.ID	STAGE	ATTEMPT_NO	STATUS	RESULT
101	PREPARATION	1	DONE	—
102	SORTIR	1	DONE	—
103	PRODUCTION	1	`VOIDED`	—
104	QC	1	DONE	`REJECTED`
105	PRODUCTION	**2**	DONE	—
106	QC	**2**	DONE	`PASSED`

Penjelasan:

- Saat reject, baris tahap `return_to_stage` s/d QC pada attempt lama ditandai **VOIDED** (kecuali baris QC yang tetap `DONE` dengan `result = REJECTED` sebagai bukti).
- `brand_units.current_attempt_no` naik menjadi 2, `rework_count` naik 1, `status = REWORK`.
- Baris `brand_qc_rejections.resolved_at` diisi saat attempt 2 lolos QC.
- Seluruh jejak (termasuk durasi percobaan gagal) tetap tersimpan untuk analisis kualitas.

4.6 Event Log — Contoh Urutan

EVENT_TYPE	STAGE	ATTEMPT	FROM → TO	ACTOR
STARTED	PREPARATION	1	WAITING → IN_PROGRESS	Budi
COMPLETED	PREPARATION	1	IN_PROGRESS → WAITING	Budi
STARTED	PRODUCTION	1	WAITING → IN_PROGRESS	Andi
PAUSED	PRODUCTION	1	IN_PROGRESS → PAUSED	Andi (`WAIT_GLUE_DRY`)
RESUMED	PRODUCTION	1	PAUSED → IN_PROGRESS	Andi
QC_REJECTED	QC	1	IN_PROGRESS → REWORK	Sari (`GLUE_WEAK`)
REWORK_STARTED	PRODUCTION	2	REWORK → IN_PROGRESS	Andi
QC_PASSED	QC	2	IN_PROGRESS → FINISHED	Sari
DISPATCHED	—	—	FINISHED → DISPATCHED	Admin

---

5. Counter Agregat Batch (Denormalisasi)

5.1 Kolom yang Dipelihara

`brand_orders.units_waiting`, `units_in_progress`, `units_rework`, `units_finished`, `units_dispatched`, `progress_pct`, `stage_counts`.

Invariant yang harus selalu benar:

```
total_units = units_waiting + units_in_progress + units_rework + units_finished +
units_dispatched
```

(`PAUSED` dihitung sebagai `units_in_progress`.)

5.2 Mekanisme Update (di dalam satu transaksi)

```
DB::transaction(function () use ($unit, $event) {
    $order = BrandOrder::lockForUpdate()->find($unit->brand_order_id);
    $unit  = BrandUnit::lockForUpdate()->find($unit->id);

    $from = $unit->status;
    // 1. ubah brand_unit_progress & brand_units
    // 2. tulis brand_unit_events
    // 3. geser counter: decrement $from, increment $to
    // 4. hitung ulang progress_pct & stage_counts
    // 5. evaluasi status batch (lihat 5.3)
});
```

5.3 Otomatisasi Status Batch

KONDISI	STATUS BATCH
Batch dibuat & unit ter-generate	`INBOUND`
Unit pertama dimulai	`IN_PRODUCTION` (isi `production_started_at`)
`units_finished + units_dispatched = total_units` dan `units_dispatched = 0`	`FINISHED` (isi `finished_at`)
`0 < units_dispatched < total_units`	`PARTIALLY_DISPATCHED`
`units_dispatched = total_units`	`DISPATCHED` (isi `dispatched_at`)

5.4 Rekonsiliasi

Command `php artisan brand:reconcile {batch?}`:

- Menghitung ulang seluruh counter dari `brand_units` (`GROUP BY status`) dan `brand_unit_progress`.
- Jika selisih, perbaiki counter dan catat ke log + isi `counters_synced_at`.
- Dijadwalkan harian (scheduler) sebagai jaring pengaman.

---

6. Tracking Publik untuk Brand

- URL: `/brand-report/{public_token}` — read-only, tanpa login, token 64 karakter acak (`Str::random(64)`), dapat dinonaktifkan via `public_enabled`.
- Data yang ditampilkan:
  1. Header: logo & nama brand, nomor batch, PO, deadline.
  2. Progress bar `progress_pct` + `units_finished/total_units`.
  3. Rincian per model/ukuran (`brand_order_items` + status unit).
  4. Distribusi per tahap (`stage_counts`) tanpa nama teknisi.
  5. Timeline `brand_batch_updates` (hanya `is_public = 1`).
  6. Riwayat surat jalan (nomor & tanggal, jumlah unit).
  7. Galeri foto per unit (`brand_unit_photos` dengan `is_public = 1`; default hanya `BEFORE` dan `AFTER` attempt terakhir).
- **Tidak ditampilkan:** nama teknisi, durasi kerja internal, alasan reject mendetail, foto `REJECT` / `PROCESS` (kecuali ditandai publik), catatan internal.

---

7. Dispatch Parsial

1. Admin memilih unit berstatus `FINISHED` (centang manual atau "Pilih semua siap kirim").
2. Sistem membuat `brand_dispatches` (nomor `SJ-BR-YYMM-DD-XXXX`) dan `brand_dispatch_items`.
3. Unit → `DISPATCHED`, `dispatched_at` diisi, event `DISPATCHED` dicatat.
4. Counter: `units_finished -n`, `units_dispatched +n`; status batch dievaluasi (aturan 5.3).

5. Constraint `UNIQUE (brand_unit_id)` pada `brand_dispatch_items` mencegah unit terkirim dua kali.
6. Cetak Surat Jalan memuat daftar unit (kode, model, ukuran), total, penerima, kurir, plat kendaraan, dan tanda tangan.

---

8. Penomoran

OBJEK	FORMAT	CONTOH
Batch	`BR-YYMM-DD-XXXX`	`BR-2610-08-0001`
Unit	`{batch}-{NN}` flat	`BR-2610-08-0001-12`
Surat Jalan	`SJ-BR-YYMM-DD-XXXX`	`SJ-BR-2610-12-0001`

Untuk batch > 99 unit, suffix otomatis melebar menjadi 3 digit (`-001` .. `-250`) agar urut leksikografis tetap benar.

---

9. Perbandingan Konsep Lama vs Baru

ASPEK	DOKUMEN LAMA (V1.0)	KONSEP BARU
Relasi ke `work_orders`	1 unit = 1 `work_orders` + kolom `brand_order_id`	**Tidak ada** relasi sama sekali
Perubahan tabel eksisting	+1 kolom di `work_orders`	**0**
Jumlah tabel baru	2	12
Penyimpanan tracking	Kolom stasiun di `work_orders` (`prep_*` , `prod_*` , `qc_*`)	`brand_unit_progress` + `brand_unit_events`
Riwayat rework	Tidak tersedia (kolom ditimpa)	Penuh (`attempt_no` , `VOIDED` , `brand_qc_rejections`)
Jeda & net time	Mengikuti trait `HasStationTracking`	Dihitung di `brand_unit_progress`
Isolasi dari CS	Disaring via status `BRAND_FINISHED`	Otomatis (tabel terpisah)
Progress batch	Hitung dari status `work_orders`	Counter denormalisasi + rekonsiliasi
Akses brand	Tidak ada	Public link bertoken
Dispatch	Satu kali, wajib 100%	Parsial per unit
Scanner & UI mobile	Memakai stasiun eksisting	Dibuat baru (`/m/brand/...`)
Foto unit	Tidak ada	`brand_unit_photos` (BEFORE/PROCESS/AFTER/REJECT, wajib di titik kritis)

---

10. Dampak & Pekerjaan Lanjutan

[!WARNING] Karena brand tidak lagi berada di `work_orders`, hal-hal berikut **tidak otomatis tersedia** dan perlu dibangun/diputuskan:

AREA	DAMPAK	KEPUTUSAN DIBUTUHKAN
Scanner & halaman teknisi	Perlu halaman mobile khusus brand	Reuse pola `ProductionControlling` & `SpkTracker`
Cetak SPK Induk & stiker barcode	Template baru (tema Amber `#d97706`)	Ukuran stiker & jumlah per lembar
Laporan/dashboard operasional	Tidak membaca data brand	Perlu dashboard brand sendiri
Penagihan & harga	Skema belum memuat harga/invoice	**Ditunda** — konsep invoice B2B belum ada; dirancang terpisah kemudian
Sinkronisasi Google Sheets	Perlu endpoint baru `sync_brand_*.php`	Format satu baris per unit (dengan `unit_id`)
Hak akses	Permission baru modul brand	Matriks role (Admin, Kepala Produksi, Teknisi, QC)
Migrasi data lama	—	**Selesai:** tidak ada data brand di `work_orders`, tidak perlu migrasi
Foto before/after per unit	Ditangani `brand_unit_photos`	**Selesai:** foto per unit per tahap, wajib di titik kritis (Bab 3.11)
Kamera & unggah di mobile	Halaman teknisi perlu input kamera + kompresi	Batas ukuran file & kompres di sisi klien atau server

---

11. Rencana Implementasi Migration (Urutan)

1. `create_brand_partners_table`
2. `create_brand_orders_table`
3. `create_brand_order_items_table`
4. `create_brand_units_table`
5. `create_brand_unit_progress_table`
6. `create_brand_unit_events_table`
7. `create_brand_qc_rejections_table`
8. `create_brand_batch_updates_table`
9. `create_brand_dispatches_and_items_tables`
10. `create_brand_number_sequences_table`
11. `create_brand_unit_photos_table` (setelah `brand_qc_rejections` agar FK `rejection_id` valid)

Lapisan aplikasi: Model + Enum (`BrandStage`, `BrandUnitStatus`, `BrandOrderStatus`, `BrandPhotoType`) → `BrandNumberService` → `BrandBatchService` (generate unit atomik) → `BrandPhotoService` (unggah, resize, thumbnail) → `BrandTrackingService` (mulai/jeda/lanjut/selesai/QC/rework + validasi foto wajib + counter) → `BrandDispatchService` → Livewire (admin, mobile, publik) → command `brand:reconcile` → feature test (alur normal, jeda, rework, foto wajib, dispatch parsial, konsistensi counter).